Cardisoft: Coronary Artery Disease Risk Estimation and 3D Visualization

Project Evaluation Report

***Abstract*—Cardisoft** is a research prototype that estimates coronary artery disease (CAD) and vessel-specific stenosis probabilities from structured clinical data. It combines **four tabular classifiers** with an interactive heart dashboard and patient-level feature explanations. Evaluation on the project's held-out test partitions shows the strongest discrimination for CAD and the left anterior descending artery (LAD); sensitivity is substantially lower for the left circumflex (LCX) and right coronary artery (RCA). The 3D display maps vessel probabilities to illustrative coronary routes, not to patient-specific lesion locations. Results are limited by the small, single-source cohort, uncalibrated serving probabilities, and absence of external validation. The system is a research and educational prototype, **not a diagnostic device**.

***Availability*—**Source code: `https://github.com/shavipal567/Cardisoft`. Live demo: `https://cardisoft.vercel.app/`.

***Index Terms*—**coronary artery disease, stenosis prediction, tabular machine learning, SHAP explanations, 3D visualization, Z-Alizadeh Sani dataset.

1. System Overview

CAD is conventionally confirmed by invasive angiography, which is costly and carries procedural risk [1, 2]; earlier work has therefore asked how well routine clinical data predict CAD and per-artery stenosis [3]. The system uses the **303-row Z-Alizadeh Sani CAD dataset** [4] (`dataset/dataset.csv`), comprising **54 clinical inputs and four binary targets**: CAD, LAD, LCX, and RCA. After removing four target-related columns and one constant field, the data undergoes category normalization and fitted preprocessing. A shared Python pipeline serves predictions through FastAPI [5] to the dashboard (Fig. 1). The dashboard displays vessel probabilities, risk estimates, feature explanations, and an interactive 3D heart with selectable LAD, LCX, and RCA reference routes. These routes are *illustrative*, *not patient-specific lesion maps*. Speech, text, and report import assist data entry; raw audio and medical images are not model inputs.

2. Dataset Preprocessing

The source cohort contains **303 records and 59 columns**: 54 permitted predictors, four diagnosis labels, and one constant exertional-chest-pain field. The targets are Cath-derived CAD and LAD, LCX, and RCA stenosis. Their positive/negative counts are CAD 216/87, LAD 177/126, LCX 119/184, and RCA 114/189. All four targets and the constant field are excluded from predictors. The inputs include 33 numeric or binary-valued fields, 17 Y/N categorical fields, and four nominal fields (Sex, BBB, VHD, Region RWMA). Fig. 2 shows the marginal distributions of selected features.

Cleaning maps the dataset typo `Fmale` to `Female`; undocumented Region RWMA codes are treated as categories rather than an assumed severity scale. Numeric inputs use median imputation and RobustScaler [6]; categoricals use most-frequent imputation. The one-hot track handles unseen categories with an all-zero encoded block. The native CatBoost/LightGBM track [7, 8] uses integer category codes and an unknown-level sentinel. No automatic outlier clipping is applied. These transforms are fitted within each training fold and saved with the final estimator, *preventing validation records from influencing* imputation, scaling, or category vocabularies.

Training also supports ratios (LDL/HDL, TG/HDL, Neut/Lymph), two clinical finding counts, and $\log(1+x)$ transforms for TG, FBS, and ESR. Ratios and counts augment the numeric inputs; log transforms replace their original lab values. Invalid denominators and non-finite operands propagate as missing before imputation. Derived values are computed inside the fitted preprocessor, so the API still accepts exactly 54 raw fields. These feature options are available to new training runs; checked-in serving artifacts and the reported metrics use *baseline inputs*.

3. Model Architecture

Four independent binary estimators produce CAD, LAD, LCX, and RCA probabilities. Training compares **32 configurations per target** from Random Forest [9], XGBoost [10], HistGradientBoosting, LightGBM, CatBoost, stacking, and soft voting. **Mean ROC–AUC** [11] over ten repeats of five-fold cross-validation [12] selects a winner, with mean F1 as the tie-breaker. Class weighting addresses imbalance [13]. The saved serving models are **CatBoost for CAD** (400 iterations, depth 4, learning rate 0.05, $L_2 = 3$, balanced weights) and **Random Forest for all vessels** (400 trees; minimum leaf size 2; RCA depth 8, LAD/LCX unrestricted depth). The CAD transformer produces 54 native columns; vessel transformers produce 81 one-hot columns.

The persisted bundle contains estimators, preprocessors, metadata, feature schemas, and reference data for explanations. FastAPI loads the frozen artifacts at inference time; it does not fit models during requests. All models return a two-column probability array ordered by classes 0 and 1. A

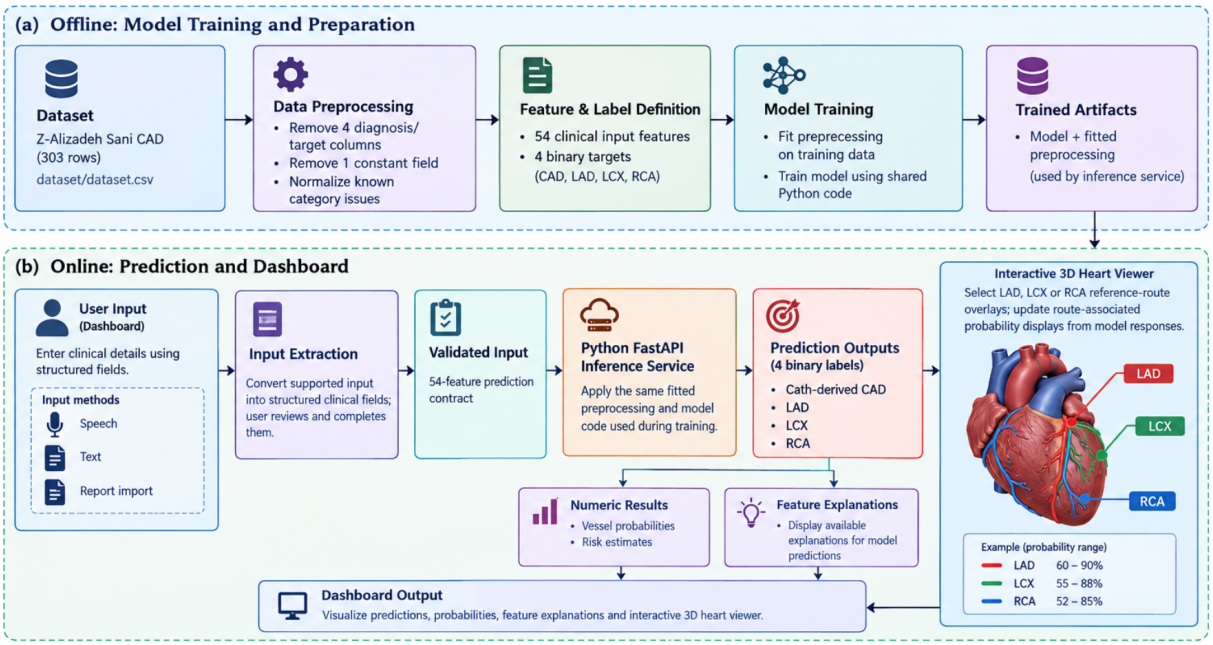


Figure 1. Overview of the proposed CAD prediction system, illustrating dataset preprocessing, machine learning inference, FastAPI integration, and interactive dashboard visualization.

saved threshold of 0.5 determines the displayed binary classification. Independent heads are simple to maintain but do not explicitly model vessel co-occurrence. A separate nested CAD experiment evaluates training-only inner selection and outer-fold performance; it is not the serving bundle. **Deployed probabilities remain uncalibrated**.

4. Predictive Performance

4.1 Validation and Results

Saved serving models were evaluated on target-specific, **stratified 80/20 partitions** (242 development and 61 held-out records per target; random seed 42). Model selection used repeated five-fold cross-validation on the development set. Preprocessing is fitted within folds, and held-out observations are not used for fitting. Table 1 reports the current saved-artifact test results at the saved **0.5 decision threshold**.

CAD and LAD have stronger held-out discrimination in this cohort: the models flag 40 of 43 CAD-positive and 31 of 36 LAD-positive records. The cost is false positives, 6 of 18 CAD-negative (specificity 0.667) and 8 of 25 LAD-negative records (0.680); the CAD specificity rests on only 18 negatives and is uncertain. Fig. 4 is a training diagnostic, not a held-out measurement. Its diverging training and validation loss indicates the onset of overfitting on this small cohort and reinforces that the held-out figures should be read cautiously. Each test partition is small and comes from the same source as the development data; these measurements do not establish performance in another hospital or population.

4.2 LCX and RCA Sensitivity

What the numbers show. At the fixed 0.5 threshold, recall is 0.458 for LCX (11/24) and 0.391 for RCA (9/23), so the models miss 13 of 24 LCX-positive (54%) and 14 of 23 RCA-positive (61%) held-out records (Fig. 3). Specificity is much higher (0.811 for LCX, 0.842 for RCA), which places the operating point at the conservative end: the models rarely flag a vessel, and when they do, roughly four in ten flags are false alarms (7 of 18 LCX and 6 of 15 RCA flagged records; precision 0.611 and 0.600). Accuracy of 0.672 for both vessels exceeds the majority-class baselines (0.607 and 0.623) by only 0.05–0.07. A classifier that always predicted "no stenosis" would be correct on 37 of 61 LCX and 38 of 61 RCA records while detecting no positives, so accuracy alone overstates these two outputs. F1 (0.524 and 0.474) is the clearest single summary of the weakness.

Ranking versus thresholding. ROC–AUC (0.711 for LCX, 0.748 for RCA) is above chance but well below CAD (0.893) and LAD (0.843), so the weakness is not only a threshold effect. Part of the low recall may nevertheless reflect the operating point, since LCX and RCA positives are the minority class (39.3% and 37.6% prevalence) and some positives may score just below 0.5. We did not evaluate other thresholds, so we cannot say how much sensitivity a lower cutoff would recover or at what cost in false positives. An AUC near 0.7 implies that no threshold can deliver both high sensitivity and high specificity for these targets.

Statistical uncertainty. With 24 and 23 held-out positives, the estimates are imprecise. Wilson 95% intervals computed from the counts span about 0.28–0.65 for LCX recall and 0.22–0.59 for RCA recall, and one additional correct or missed case shifts recall by about 0.04. The two vessels are therefore statistically indistinguishable here, and the small difference between their metrics should not be interpreted.



Figure 2. Comparative boxplot distributions of selected clinical features in the original (blue) and Z-copied (green) datasets, illustrating differences in central tendency, variability, and outlier patterns.

Table 1. Held-out test performance of the saved serving models (61 records per target, threshold 0.5).

Target	Accuracy	Precision	Recall	F1	ROC–AUC	TN/FP/FN/TP
CAD	.852	.870	.930	.899	.893	12/6/3/40
LAD	.787	.795	.861	.827	.843	17/8/5/31
LCX	.672	.611	.458	.524	.711	30/7/13/11
RCA	.672	.600	.391	.474	.748	32/6/14/9

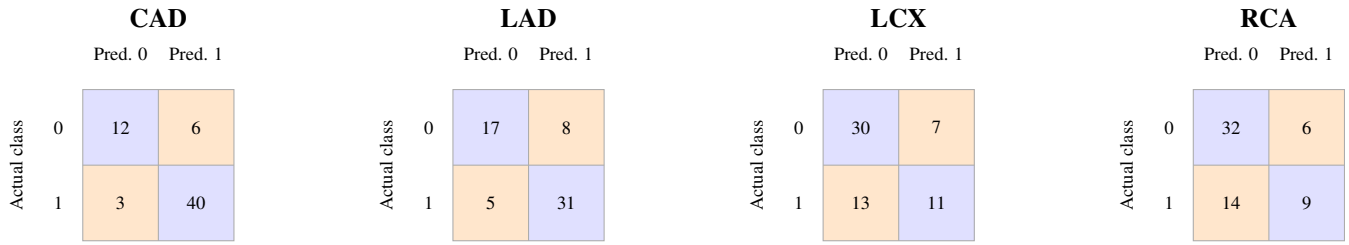


Figure 3. Held-out confusion matrices for CAD, LAD, LCX, and RCA (61 records per target; decision threshold 0.5). Rows denote actual class and columns predicted class, with class 1 indicating a positive diagnosis. Diagonal cells (blue) are correct classifications; off-diagonal cells (orange) are errors. Counts are taken directly from Table 1; the four target-specific test partitions need not contain the same records.

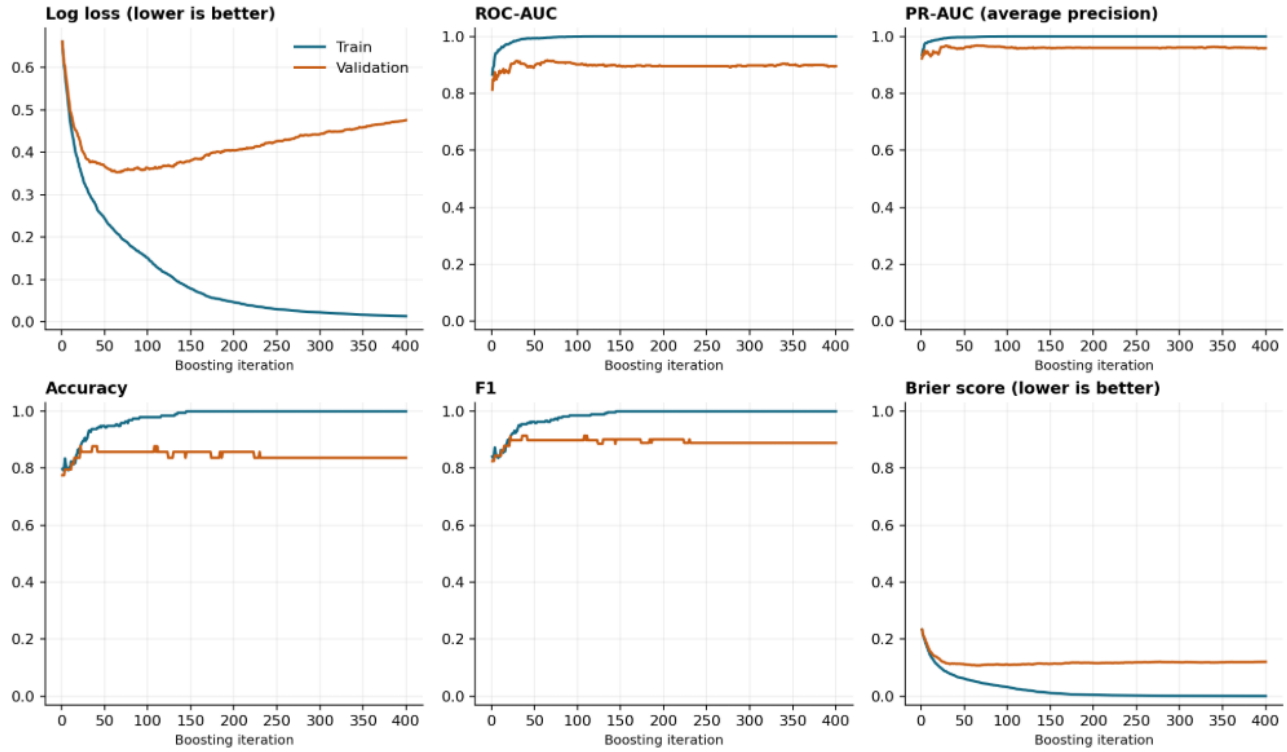


Figure 4. Training (blue) and validation (orange) performance across 400 boosting iterations, evaluated using six metrics: logarithmic loss (Log Loss), ROC–AUC, PR–AUC (Average Precision), Accuracy, F1-score, and Brier score. The declining training loss and consistently high training metrics indicate progressive model fitting, while the increasing validation loss and relatively stable validation metrics suggest the onset of overfitting. Lower values indicate better performance for Log Loss and Brier score, whereas higher values are preferred for the remaining metrics.

Possible causes (untested). Three explanations are plausible. First, there are fewer positive development examples for these vessels (roughly 95 for LCX and 91 for RCA) than for CAD and LAD (roughly 173 and 141). Second, the CAD label aggregates narrowing in any artery, so more recorded findings may carry signal for it than for a single vessel. Third, the independent heads receive no information about vessel co-occurrence. We ran no learning-curve, ablation, or threshold experiments, so none of these is confirmed, and the differing RCA depth limit should not be read as an explanation.

Implications. A missed stenosis is the costlier error wherever an output could be read as reassurance. With 3 of 43 CAD cases already missed, let alone the LCX and RCA miss rates, none of the outputs should be used to rule out disease, and the LCX and RCA outputs should be treated as weak signals. The same caution applies to the 3D display (Section 5): a low-probability color on the LCX or RCA route does not indicate that the artery is unaffected.

4.3 Nested Validation and Calibration

A separate completed CAD nested-validation experiment used ten seeds with five outer folds and inner model selection, because selecting and evaluating a model on the same data yields optimistic estimates [14, 15]. For its nested candidate, mean outer ROC–AUC was 0.9280 (SD 0.0328), sensitivity 0.9201, specificity 0.7316, and log loss 0.3137. The baseline

mean outer ROC–AUC was 0.9273 (SD 0.0304); the paired patient-bootstrap interval for the mean log-loss change was $[-0.0639, +0.0123]$, which includes zero. This experiment is useful methodological evidence but does not replace the serving metrics or establish a generalization improvement. Its protocol, partitions, and candidates differ from those of Table 1, so its higher AUC does not contradict the held-out AUC of 0.893.

Serving models remain uncalibrated: a score of 0.7 has not been shown to mean that 70% of similar patients have the condition. Post-hoc calibration [16] needs additional data and has not been applied, and the 0.5 threshold is a convention, not a value tuned to clinical costs. Probability values **must not be interpreted as clinically validated individual risk percentages**.

4.4 Assessment

The project implements stratified held-out testing, repeated CV, fold-contained preprocessing, and a more rigorous nested CAD experiment. Reporting multiple metrics and confusion counts supports assessment beyond accuracy. Key limitations are the 61-patient test size per target, one dataset source, **no independent external validation**, and legacy model-selection results that may be optimistic after extensive exploration. Vessel-specific sensitivity in particular requires improvement and prospective evaluation before clinical use.

5. 3D Visualization

5.1 Rendering and Reference Routes

The dashboard renders the heart in the browser with Three.js [17] through React Three Fiber [18]. The anatomical asset is a BodyParts3D/Anatomography heart model [19, 20] supplied as a GLB file whose mesh chunks have generic names and carry no verified LAD, LCX, or RCA segmentation. The reference routes are therefore constructed separately. Asset-specific landmarks based on normal anatomy define the LAD, LCX, and RCA paths; the registration process interpolates route samples between landmarks and projects them onto the heart surface; and each route is rendered as a tapered tube mesh with a selectable label. LAD and LCX share an illustrative left-main origin, while RCA has a separate origin. The left-main connector stays neutral in color because no model output exists for it.

After changing route landmarks or the asset, run `npm run anatomy:register` and `npm run anatomy:check` from `frontend/`. These checks verify asset fingerprint, route continuity, origin, direction, and surface contact. They test geometric consistency, *not clinical anatomical accuracy*.

5.2 Integration with Predictions and Explanations

The 3D scene is a presentation layer driven by the same API responses as the risk cards (Fig. 5). The integration proceeds as follows.

1. The dashboard submits the reviewed 54-field input to the prediction API, which returns the CAD, LAD, LCX, and RCA probabilities from the frozen models.

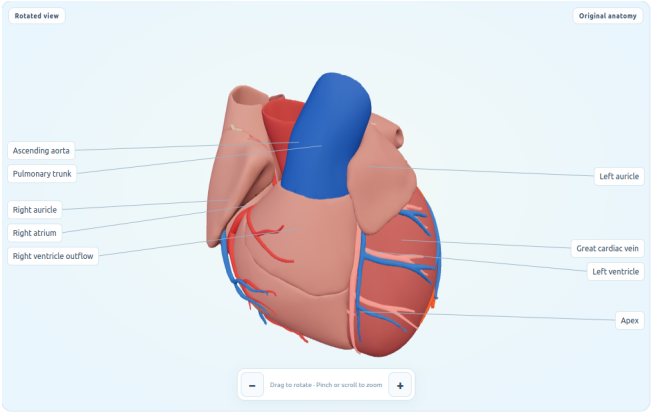


Figure 5. Interactive 3D heart visualization integrated into the CAD prediction dashboard, displaying anatomical reference structures and selectable coronary artery routes (LAD, LCX, and RCA). The interface supports anatomical exploration and contextual visualization of vessel-specific predictions. Highlighted arterial pathways are illustrative and do not represent patient-specific coronary lesions.

2. Each vessel probability is passed through a continuous color scale to give one uniform color for that vessel's route, and the numeric probability is shown on the route label. Color is therefore never the only cue. The CAD output is patient-level, so it appears on its risk card and is not assigned to a route.
3. On each new response, only route materials and labels are updated; the heart geometry is not rebuilt.
4. Selecting a vessel focuses and highlights its route and opens the inspector panel with that vessel's probability and explanation table. When the user requests an explanation, the attribution endpoint is queried for that selected target (Section 6), so the costlier computation runs on demand.

The design separates a static layer from a dynamic one. Route geometry is registered offline and checked by the scripts above, so it never changes with a prediction, and route appearance is a function of the vessel probability alone. Risk cards and explanations remain available independently of the 3D view, so the display adds spatial context without altering any model output. It also cannot make the outputs look more reliable than they are: the LCX and RCA colors carry the limited sensitivity described in Section IV-B.

5.3 Anatomical Limitations

Anatomical claims are deliberately bounded. The heart model is illustrative, and the coronary routes are manually registered reference illustrations based on normal anatomy, *not clinically validated segmentations*. The dataset contains no imaging, lesion coordinates, plaque geometry, or patient-specific anatomy. A highlighted route means only that the model's estimated probability of stenosis somewhere in that artery has a given value; it **does not locate or characterize a lesion**, and color must not be used to infer plaque burden, lesion position, or branch involvement. A continuous color scale applies one

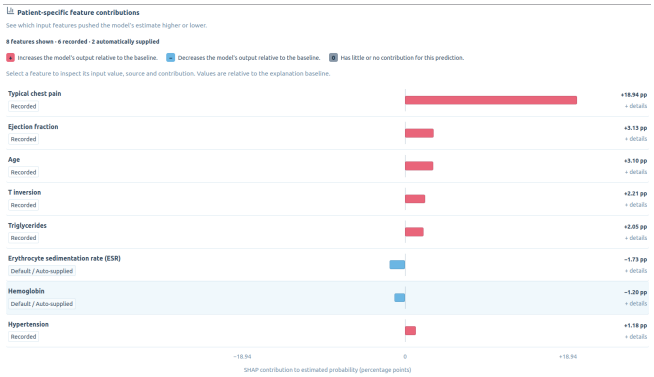


Figure 6. Patient-level Shapley attribution from the Cardisoft explanation service for one selected target. Each entry is a signed change in predicted probability relative to the reference baseline, grouped by original clinical field: positive contributions raise and negative contributions lower the model score, and undisplayed fields are collected in a residual so that baseline, contributions, and residual sum to the prediction. The attribution describes model behavior for a single record; it is approximate, depends on the reference background, and is not a causal effect.

value per vessel, so there is no local plaque or lesion map. Appearance and opacity controls, camera presets (anterior, posterior, lateral), and a responsive layout support exploration without changing these constraints.

6. Clinical Interpretability

Shapley values [21] share a joint payout among players by their average marginal contribution over all orders in which players could join. Treating features as players and the model output as the payout gives Shapley-based attribution [22]. For a selected target, the explanation service computes **approximate permutation Shapley contributions** [23] in probability space, so contributions sum to the difference between the prediction and the baseline estimate. It returns signed feature contributions, raw feature values, a baseline estimate, and a residual for contributions outside the displayed top features. Features are grouped back to their original clinical fields, including one-hot encoded categories. This gives an individual assessment a ranked, directional account of which inputs moved the model prediction relative to its reference background. The dashboard presents the explanation alongside physiological field values, with clinician and patient wording modes (Fig. 6).

These are explanations of model behavior, **not causal effects** or independent evidence of disease mechanism. Contributions depend on the model, reference background, and correlated predictors; the permutation computation is approximate. Sparse categories and the small cohort can also make explanations unstable. The report should therefore be read as a transparent account of the model's estimate rather than a clinical rationale for treatment.

7. System Integration

The frontend submits structured patient inputs to the prediction API and receives CAD and vessel probabilities, effective and defaulted input values, model identities, and available-input validation context (Fig. 7). A separate explanation endpoint supplies feature attributions for the selected output. Model information and held-out metrics are exposed through an API endpoint. The dashboard uses the returned probabilities to refresh risk cards and the 3D vessel overlays after each analysis. Form, speech/text parsing, OCR, and selectable-text PDF extraction converge on the **same reviewed 54-field structured input contract**.

The available-input accuracy panel evaluates frozen models on reconstructed held-out cohort records after substituting defaults for fields not supplied. It is a cohort-level exploratory measurement, *not confidence in an individual prediction*. Imported values require user review before analysis; import does not itself trigger model inference.

8. Technical Implementation and Availability

The implementation separates training, inference, API, and browser visualization. Model artifacts retain metadata and feature schemas; the same preprocessing logic is shared across training and serving. Evaluation outputs and experiment provenance are stored with the project, while API validation constrains inputs to the documented schema. The use of a public tabular dataset and an attributed third-party heart asset supports reproducibility and resource attribution.

Availability. The repository contains the dataset, training scripts, preprocessing, model artifacts, API services, frontend, evaluation outputs, and documentation [24]: `https://github.com/shavipal567/Cardisoft`. A live demo is deployed at [25]: `https://cardisoft.vercel.app/`. The demo is a research and educational prototype and carries all limitations stated in this report.

Reproducibility does not resolve dataset provenance or transportability. The data contain 303 rows and no verified patient identifier or acquisition timestamp, so independence of repeated patients and pre-decision availability of every field cannot be established. The heart model has no clinically labelled coronary segmentation. These constraints, and the absence of external validation, are material technical and clinical limitations.

9. Conclusion and Next Steps

Cardisoft combines a functioning tabular prediction pipeline, patient-level explanations, and interactive 3D presentation. Its strongest current evidence is for CAD and LAD discrimination on the available cohort; LCX and RCA sensitivity is limited (13 of 24 and 14 of 23 positives missed at the fixed threshold). The visualization provides spatial context but does not show patient-specific pathology. Priorities for further development are independent multi-site validation, threshold and sensitivity analysis for LCX and RCA on larger samples, calibration studies on adequate samples, provenance review, and clinically reviewed coronary segmentations if patient-specific spatial claims are pursued. Until those steps are completed, Cardisoft should be used for **research and education only**.

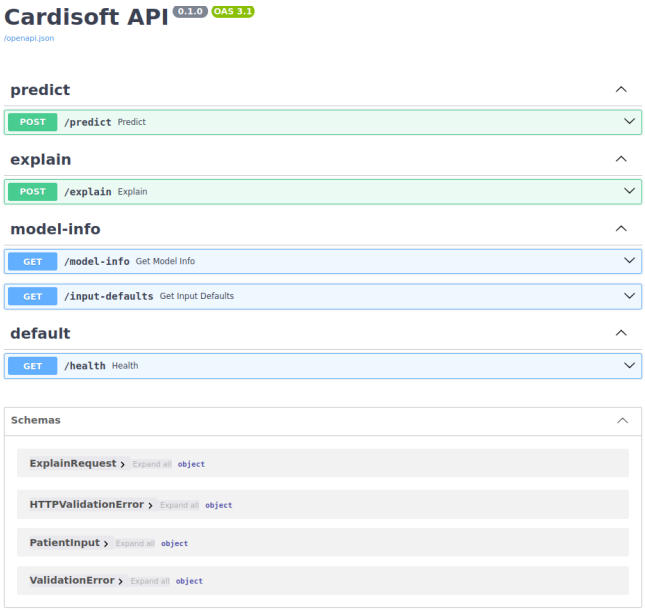


Figure 7. Cardisoft FastAPI endpoints. The prediction endpoint validates the reviewed 54-field input and returns the CAD, LAD, LCX, and RCA probabilities with effective and defaulted input values and model identities. A separate explanation endpoint returns signed attributions for the selected target on request, and a model-information endpoint exposes model metadata and held-out metrics. The endpoints serve frozen model artifacts and never fit models at request time.

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